



Wear performance of short fiber added polyamide composites produced by additive manufacturing: Combined impacts of secondary heat treatment, reinforcement type, and test force

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Abstract

In recent years, the number of studies focusing on the additive manufacturing has increased seriously to elucidate the critical points such as physical, chemical, and mechanical properties. Contrary to common trends and for the first time in the literature, this experimental endeavor aims to comprehend the combined impact of reinforcement type and heat treatment on the wear features of additively manufactured polyamide 6 (PA6) composites. As reinforcement materials, glass, and carbon fibers were utilized and all samples were created through the fused filament fabrication. Half of the samples were subjected to annealing treatment after the main production, and characterization works were performed using a fused emission scanning electron microscope, differential scanning calorimetry, and dynamic mechanic analysis. From the outcomes, it is seen that heat treatment has a positive effect on the hardness, and wear resistance of the composites. Besides, glass fiber-reinforced samples display lower friction coefficient and lower volume loss results than other samples. For all samples, secondary annealing causes a positive impact on wear endurance in most cases. On the other side, the wear mechanism of the tested samples changes with the test force level and reinforcement type. At lower test forces, abrasive wear-induced debris parts are detected on the deformed surfaces of PA6 and carbon-added samples, but this case is opposite for the highest force of 40 N.

Highlights

- Additive manufacturing was underlined as a promising way to create polymer composites with high dimensional accuracy.
- The combined effect of secondary annealing and reinforcement type on the wear resistance of 3D-printed PA6 composites was examined for the first time in the literature.
- Carbon and glass fibers were compared to explore their effect on friction coefficient and wear behavior of polymer composite samples produced via FFF technology.

- Hardness and surface roughness measurements were used to comprehend the dry sliding wear results.
- Depending on the increasing test force levels and reinforcement type, deformation, and abrasion mechanisms were analyzed.

KEYWORDS

carbon fiber, friction coefficient, fused filament fabrication, PA6 composites, wear mechanism

1 | INTRODUCTION

Additive manufacturing (AM), or in other words three-dimensional (3D) printing, is a new-generation layer-by-layer fabrication style and provides notable manufacturing flexibility to customers in comparison with traditional ways like welding, machining, plastic forming, and casting. Nowadays, owing to the rising interest in this alternative methodology, it is not difficult to foresee that AM and AM-connected techniques will be the most preferable fabrication ways in the next decades for manufacturers, engineers, scientists, and customers. When the main targets and likely obtainments are considered at a sectoral level, it is seen that the biomedical, medicine, pharmacy, food, energy, composite, aerospace, automotive, pipe, and building industries are the frontrunners.^{1–6} Considering the innovative perspectives in the industry, along with the escalating concerns on environmental friendliness, air pollution, and climate change, it is not difficult to claim that the rising trend of AM technologies will continue, and its usage area will expand. In that sense, layer-by-layer production is one of the strong alternatives to take a noteworthy share in the general manufacturing methods soon.

In recent years, with the increasing technological developments in the 3D printing area, available material portfolios have begun to diversify from polymers and metals to composites and ceramics.^{7–10} Besides, this situation is also open to widening depending on the valuable outcomes that will be attained from scholarly efforts. As for the general printing strategies, there are different approaches in the scientific archives. However, the classification of the primary system into the seven sub-categories is the most adopted way by researchers and it was also standardized by international organizations (ISO 52900 and ASTM F2792). According to this classification, the fundamental approaches can be counted as vat photo-polymerization, material extrusion, material jetting, binder jetting, powder bed fusion, sheet lamination, and directed energy deposition.^{11–13}

Considering the literature works concentrating on the AM, it is comfortably noticed that polymer-based materials and material extrusion-based fused filament fabrication (FFF)

methods are more widespread than the other techniques. This circumstance can be attributed to the relatively low melting point of polymers, sufficient fluidity of the filaments, and cost-effective/practicable system requirements of FFF devices. Thanks to these affirmative features, several polymer parts printed via FFF are used frequently in medical implants,¹⁴ orthopedical equipment,¹⁵ drug delivery systems,¹⁶ rapid prototyping efforts,¹⁷ and automotive components.¹⁸ If the latest trends shared in the literature are scanned meticulously, as major filament materials, acrylonitrile butadiene styrene (ABS), polylactic acid (PLA), polyether ether ketone (PEEK), polyethylene terephthalate glycol (PETG), and polyamide (PA) can be distinguished comfortably. As such, composite studies proceed also substantially with these polymers and usually reinforcement materials alter. Lately, the number of polymer composite investigations has begun to increase to a large extent, and the accumulated knowledge on FFF has been combined with the composite area. At this point, primary endeavors are on the mechanical properties and their relationship with the printing parameters. In this context, lots of significant surveys discussing tensile, flexural, impact and ductility responses are present in different technical papers.^{19–23} However, there are limited investigations on the wear and friction behaviors of FFF-based polymer composites even though this subject is nonignorable for the system components working with contact/friction with other parts such as implant joints and gears. This case has started to change slightly in recent years and the area has gained interest from certain research groups. In the previous studies, Rajkumar et al.²⁴ used FFF technique and pointed out that increasing the weight percentage of silicon (up to 7%) enhanced the wear endurance of the 3D-printed PLA samples and pure PLA samples showed more abrasive wear tendency compared to their composite versions. The researchers found that friction coefficient values dropped to 0.11 with the metal addition from 0.24 and PLA samples with 7% Si particles had $1.8 \times 10^{-3} \text{ mm}^3/\text{Nm}$ specific wear rate while the plain PLA samples possessed $5.3 \times 10^{-3} \text{ mm}^3/\text{Nm}$. Haq et al.²⁵ produced carbon fiber (CF) reinforced PETG parts via FFF to analyze the tribological features and

pointed out that composite samples with different layer thicknesses exhibited lower friction coefficient levels between 0.1 and 0.25. Keshavamurthy et al.²⁶ utilized FFF technology and enounced that graphite addition to ABS matrix dropped the total wear loss from 400 μm to 50 μm due to the solid lubricant effect of graphite. The researchers also showed that friction coefficient levels were prone to go down with the addition of graphite (0.2 for composites and 0.8 for plain samples). Similarly, Przekop et al.²⁷ tried graphite modification in the PLA matrix for FFF printing and indicated that the lowest wear loss of 11 $\mu\text{m}/\text{km}$ belonged to the highest graphite weight percentage of 10%. Besides, the project group propounded that friction coefficient results fluctuated between 0.40 and 0.45 for all specimens. In another work, Agarwal et al.²⁸ proved the positive influence of graphene contribution to PLA matrix to improve the creep and wear resistance of 3D-printed specimens. They asserted that volume loss values diminished from 4 mm^3 to 3.5 mm^3 following the graphene mixing and average friction coefficient values went down 0.2. Singh et al.²⁹ obtained Al_2O_3 -filled Nylon6 filaments via a single screw extrusion technique for the FFF method and emphasized that dual particle size added specimens had superior wear properties than the others. Prusinowski et al.³⁰ analyzed the effect of the tribological test environment on the wear properties of additively manufactured continuous CF-reinforced ABS polymer composites and revealed that kinematic friction coefficient values (detected between 0.50 and 0.25) were lower in the water environment than those of the air environment. The intensity of mass wear of composites was higher in the case of dry conditions than in the water and changed in the range of 1.25×10^5 – 2.65×10^5 $\mu\text{g}/\text{m} \cdot \text{m}^2$. Soundararajan et al.³¹ put forth that the wear behaviors of TiO_2 -reinforced PA6 composites improved with the rising weight ratio of oxide particles, and the best performance was reached for the samples having 30% TiO_2 (0.8 mm^3/Nm wear rate for 20 N test force). The survey team also noted that the worst performance was detected for the plain PA6 samples (4.82 mm^3/Nm wear rate for 20 N test force). From these valuable investigations, it can be deduced that the change in the friction coefficient values not only affects the wear rates but also determines the deformation/damage mechanism of the tested samples. In general, increasing friction coefficient levels lead to more contact surfaces under the test force and trigger the increase in the wear amount, as well as the abrasive wear. However, as the friction coefficient decreases as happened in the self-lubricant graphene and carbon added polymers, adhesive style deformation becomes apparent in the contact surfaces and the initial contact condition starts to alter from composite/abrasive to composite/composite in some sections of the wear track.

In the point of perfect structural consistency and superior mechanical responses, it is known that heat treatment, or in other words annealing, is an effective and useful way for frequently utilized filament polymers.^{32,33} In particular with polymeric materials produced via additive manufacturing, secondary heat treating not only closes the layer gaps that occur during the layer-by-layer printing, but it also prompts the increase in crystallization degrees of the main plastic chains.³² This case affects directly the mechanical properties of the heat-treated samples. If the literature efforts are analyzed meticulously, the topic of heat-treating optimization and annealing-based tribological behaviors of the printed parts needs to be enlightened scientifically due to the limited number of studies in this investigation area. This background can be adapted to the wear/friction features of the 3D-printed composites and the findings can pave the way for future possibilities for longer product life. Therefore, differently from the previous literature efforts, the combined effect of the heat treatment and reinforcement type (CF, and glass fiber [GF]) on the dry sliding wear performance of the FFF-based PA6 composites was examined elaboratively in this article. The main goal of this experimental work is to explore the impact of the post-production annealing process on the hardness, crystallinity, friction coefficient, and wear resistance of PA6 composites depending on the two different fiber materials and test force levels. In addition, the detection of the wear mechanisms is targeted by means of detailed macroscopic observations, microscopic inspections, and contact surface analyses. Together with the interpretation and discussion of the obtained results in this effort, preference for the annealed composite parts possessing longer service life in harsh conditions can be encouraged.

2 | MATERIALS AND METHODOLOGY

2.1 | Composites constituents

As a matrix polymer, PA6 was selected due to its chemical resistance,³⁴ low density,³⁵ good printability with FFF,²⁹ availability for fiber reinforcement,³⁶ and impact strength.³⁴ Typically, PA6 can be formed through the ring-opening polymerization of caprolactam, which includes carbon atoms (six atoms) in its repetitive body. By reason of the forcible attraction between each molecule chain and the polarity of the amide structure, this polymer group exhibits chemical consistency along with mechanical advantages and is the most popular amide in terms of commercial activities.³⁴ On the other side, as

stated also in some studies,³⁶ pure PA grades are prone to be warped or distorted following the 3D printing operation and this situation entails dimensional/geometrical inaccuracy. Hence, as happened in this work, pure PA grades often are combined with rigid ceramic fibers like carbon and glass to develop better mechanical responses. Additionally, unwanted warpage and geometrical divergence cases can be diminished via composite structures.

Glass and carbon fibers were used in this work as reinforcement materials since these materials have low expansion rates, high elastic modulus, good abrasion resistance, and high strength.³⁵ Thanks to these hard and wear resistance fibers, the wear performance of the PA6 composites can be escalated as a result of the strong phase and layer bonding provided by the FFF technique. In this work, using secondary heat-treating, mechanical responses of the matrix polymer were aimed to be increased with the modified cross chains in the molecular structure of PA6, PA6/CF, and PA6/GF samples. Also, applying annealing has the potential to decrease the layer distance to minimize the production gaps and fiber/matrix interface volumes. Thermal activities taking place in the vicinity of the layer gaps and the contact zones stimulate diffusion-oriented surface merging cases.

All pure PA6 and composite filaments were procured from the Teknodizayn 3D Industry and Trade Limited

Company (Ankara, Turkey). For composites, Dynamide®CF and Dynamide®GF models were used. In Table 1, some physical and mechanical properties of the printed filaments can be seen according to the technical information taken from the standardized tests that were conducted and reported by the supplier firm.³⁷

2.2 | 3D printing strategy

In reviewing the literature, it can be noted that the correct combination of 3D printing parameters such as infill rate, infill type, and layer height, play a significant role in the wear properties of additively produced parts. For this work, the FFF printing steps have been stated in detail in Figure 1. If the process flow is glanced at, first, a 3D sketch was created in AutoCAD 2022 design software according to the related test standard (10 mm diameter and 20 mm height), and then exported to Ultimaker Cura 5.0.0 software for slicing operation. Afterward, the assignment of 3D printing parameters, which were reflected in Table 2, was made via the same slicing program. Following this procedure, the resulting g-code file was transferred to the LOOP model 3D printer, which operates on the principles of FFF technology. The transferred g-code file was activated in the 3D printer along with the

TABLE 1 Some physical and mechanical features of the filament materials.³⁷

Materials	Density (g/cm ³)	Melting point (°C)	Tensile strength (MPa)	Young Modulus (MPa)	Elongation at break (%)
Pure PA6	1.15	227	56	2166	4.3
PA6/CF	1.17	219	105	7500	3
PA6/GF	1.2	215	85	4400	3.5

Abbreviations: CF, carbon fiber; GF, glass fiber; PA6, polyamide 6.

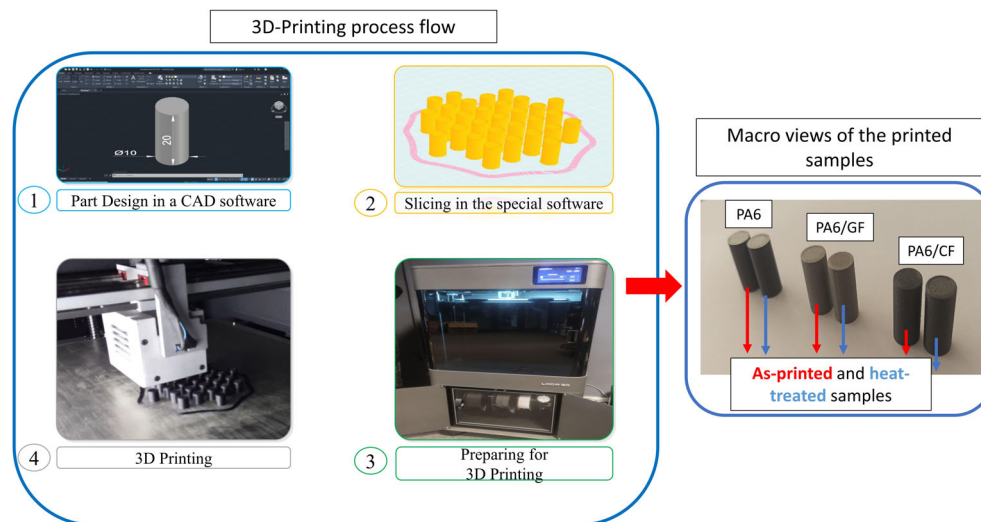


FIGURE 1 Flowchart of the 3D printing of the samples.

TABLE 2 Manufacturing parameters during 3D printing of the samples.

Parameter	PA6/PA6-CF/PA6-GF
Layer height (mm)	0.1
Infill rate (%)	100
Infill type	Linear
Printing speed (mm/s)	40
Fan speed (%)	100
Building orientation	Vertical
Raster angle (°)	0/90
Support structure	None
Adhesion type	Skirt
Number of contour	3
Nozzle temperature (°C)	260
Building platform temperature (°C)	50

Abbreviations: CF, carbon fiber; GF, glass fiber; PA6, polyamide 6.

preparation of the used filaments, and finally, the 3D printing process was completed after the production of all samples.

When the printing operation was over, the first base-line layer structure was separated from the cylindrical specimens. Macro observations were held to check whether the major surface or body defects were present in the products. From these inspections, there were no obvious major flaws in the solid bodies of the samples. Further, geometrical controls were performed to ascertain the dimensional accuracy of the printed samples. As a consequence of the dimensional measurements that were done via a digital caliper, the average diameter and height values of the samples were 9.953 mm and 19.932 mm with the accuracy values of 99.5 and 99.6% respectively.

2.3 | Characterization works and experimental details

To ascertain the production quality, printed samples were subjected to visual and microscopic inspections. In the microanalysis, a fused emission scanning electron microscope (FESEM) (Zeiss Gemini) and an optical microscope (Nikon ECLIPSE LV150NL) were used. In addition, melting temperatures and glass transition temperatures of all types of samples were found via differential scanning calorimeter (DSC) (DSC 131 EVO) and dynamic mechanic analyses (DMA) (DMA 242 E Artemis).

For heat treatment processes, an electrical resistance close cabinet furnace (Protherm HLF 100) was utilized. Applied annealing temperature values were determined based on the DSC and DMA analyses for PA6, PA6/CF,

and PA6/GF respectively. The annealing temperature values were adjusted between the melting points and glass transition points of the samples and the temperature difference between the glass transition and annealing temperature was almost taken same for all types of samples. Total annealing duration of 40 minutes was chosen according to the previous literature works.^{38,39}

Hardness tests were performed via a Shore D hardness tester (Duroironic model 1000), and each measurement set was repeated five times. As for the tribological tests, dry sliding pin-on-disc test equipment (Turkyus Podwt wear test machine) was used and four different force values were selected (5, 10, 20, and 40 N). The real view of the friction test setup can be glanced at in Figure 2.

During all tests carried out for 2 minutes at a speed of 2 m/s, friction force data were collected using Esit Data Logger V1.1.8 software and the coefficient of friction levels of the specimens were obtained. Mass change measurements were followed both before and after each test with a precision balance (Radwag AS/220/C/2). Volume loss and coefficient of friction values were calculated according to Equation 1 and Equation 2, respectively. Besides, surface roughness analyses were carried out with a surface profilometer (Mahr MarSurf PS1) in order to comprehend the wear mechanism and the nature of the deformation.

$$\text{Coefficient of Friction} = \frac{\text{Frictional Force (N)}}{\text{Normal Force (N)}}. \quad (1)$$

$$\text{Volume Loss} = \frac{\text{Mass loss (g)}}{\text{Density (g/mm}^3\text{)}}. \quad (2)$$

3 | RESULTS AND DISCUSSION

3.1 | Dimensional accuracy and surface roughness

It is known from the literature efforts that 3D printing parameters such as layer thickness, extrusion temperature and part orientation significantly affect the dimensional accuracy of 3D printed objects.^{10,40–42} In this study, the heights and diameters of the parts produced by additive manufacturing were measured with a digital caliper and the measurement results were shared in Figure 3A below. As a result of the measurements, PA6 parts were produced with 99.04% (SD:0.038) and 98.40% (SD:0.043) accuracy in height and diameter, respectively. In addition, it was determined that PA6-CF and PA6-GF parts can be produced with higher accuracy in both

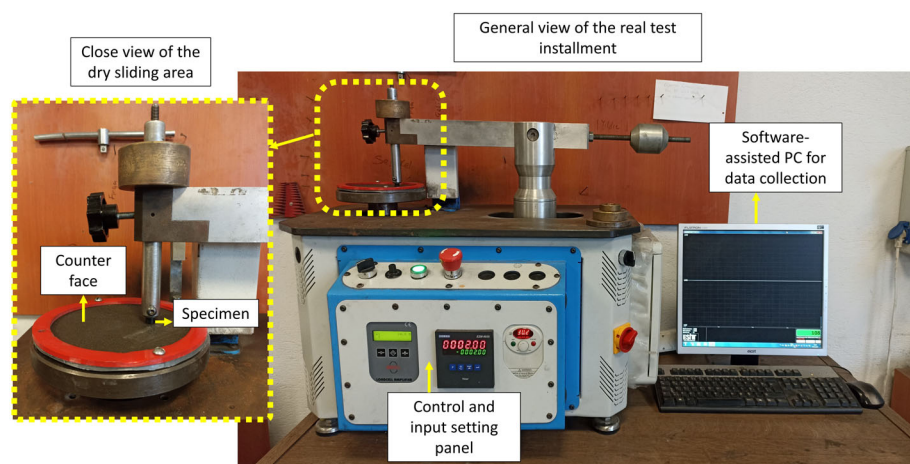


FIGURE 2 Real images of the dry sliding test setup.

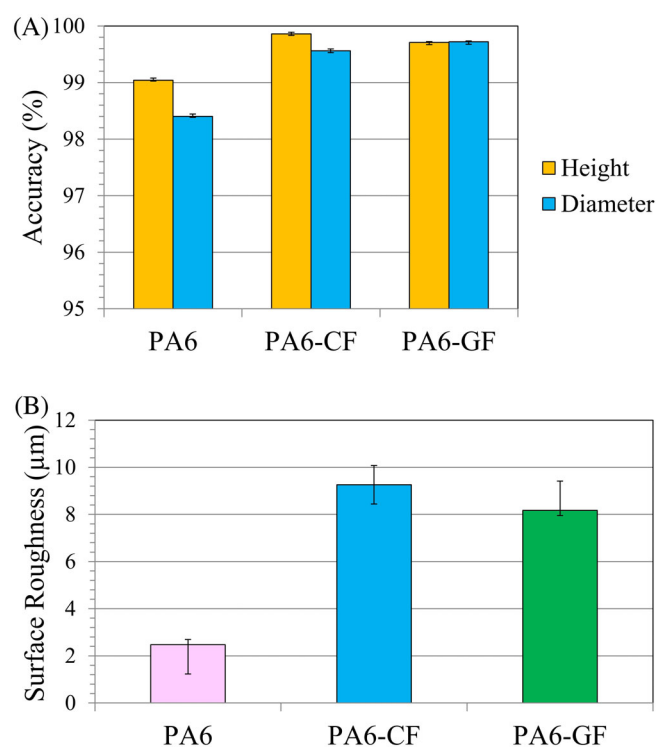


FIGURE 3 Dimensional accuracy (A) and surface roughness (B) results of the produced samples.

height and diameter compared to PA6 parts. Moreover, the accuracy of PA6-CF parts in height is 99.86% (SD:0.027), while this value is 99.71% (SD:0.014) for PA6-GF parts. Finally, in terms of diameters, the highest accuracy was observed in PA6-GF with 99.72% (SD:0.016), followed by PA6-CF with 99.56% (SD:0.032) and PA6 with 98.40% (SD:0.043). The findings indicate that reinforcement addition leads not only to increase mechanical performance but also results in higher dimensional accuracy on the products due to the thermal stability of the fibers. Obtained results point out the

precision ability of the additive manufacturing and similar levels were also reported by other research teams. Hanon et al.⁴³ conducted a study on the effects of 3D printing parameters and filament colors on the dimensional accuracy of cylindrical parts produced by FFF. As a finding of their study, they reported that the accuracy in height ranged from 98.36% to 99.72%. In addition, the diameter accuracy was also calculated and expressed as a minimum of 98.45% and a maximum of 99.63%.

Figure 3B shows the surface roughness values of the 3D printed samples. The highest average surface roughness value was obtained in PA6-CF sample with 9.25 μm (SD:0.81 μm). On the other side, the minimum average surface roughness value of 2.47 μm (SD:0.21 μm) was measured on the PA6 specimen. Furthermore, the surface quality of PA6-GF specimens was between PA6 and PA6-CF with 8.17 μm (SD:1.24 μm).

Optical microscope images were also taken to evaluate the surface roughness values of the samples better and the related images are demonstrated in Figure 4. Glancing at the micro images, it can be seen that the presence of carbon or glass fiber content in PA6 leads to an indented surface and this may increase the surface roughness values. Herein, the findings of this study are in agreement with previous literature performances. For instance, Mustaq et al.⁴⁴ investigated the effect of printing parameters on the surface roughness of PA6 samples by ANOVA analysis. According to the results of the study, it can be stated that the surface roughness values of PA6 samples vary between 4.81 μm and 12.86 μm. In another study, Dzienniak and Pawlik⁴⁵ compared the surface quality of PA6, PA6-CF20 (including 20% CF) and PA6-GF25 (including 25% GF) samples produced with FFF. As a result, they reported that PA6-CF20 showed the highest Ra value with 12.83 μm, followed by PA6-GF25 and PA6 with Ra values of 11.60 μm and 6.33 μm, respectively.

FIGURE 4 Microscopic images of the samples, (A) polyamide 6 (PA6), (B) PA6-carbon fiber (CF), (C) PA6-glass fiber (GF).

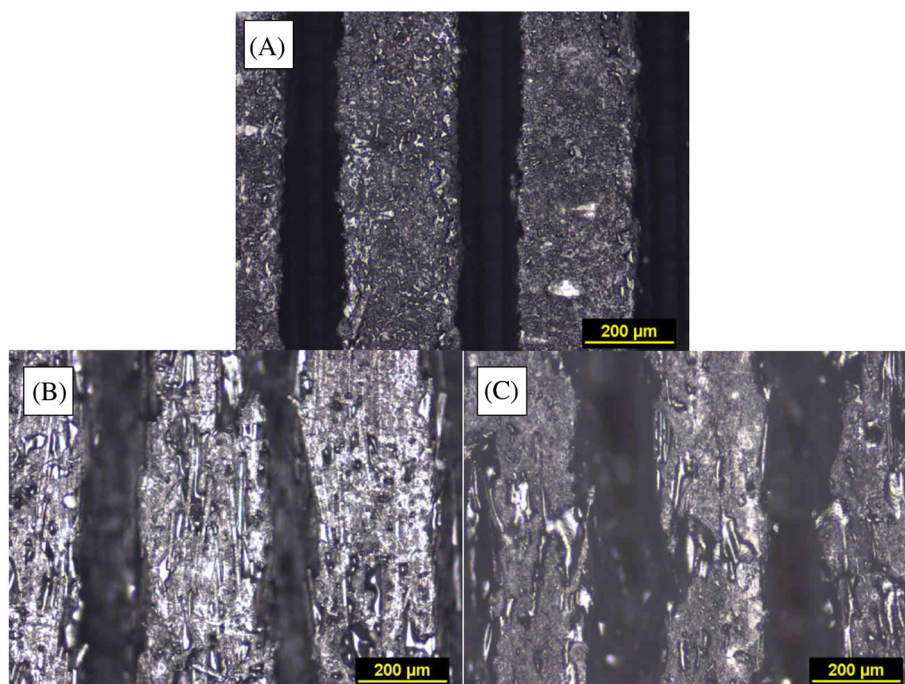
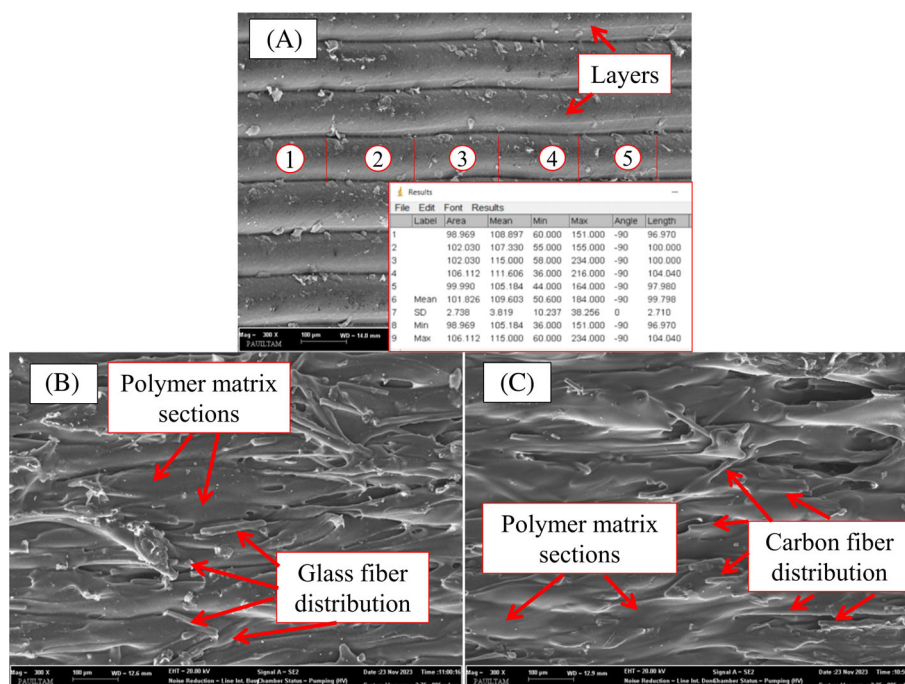


FIGURE 5 Fused emission scanning electron microscope (FESEM) images of the cross-sections of the samples; polyamide 6 (PA6) (A), PA6/glass fiber (GF) (B), and PA6/carbon fiber (CF) (C).



3.2 | Microstructural, DSC, and DMA analyses

After the printing operations and preliminary controls of the parts, scanning electron assisted microscopic inspections were carried out to examine the inner phase structure of the fabricated samples. Figure 5 indicates the FESEM images of the cross-sectional views of the samples. From these images, carbon and glass fibers in the PA6 composite system can be distinguished and their distribution circumstance in the

matrix can be seen. It can be asserted that a relatively homogeneous reinforcement distribution exists in the solid PA6 filament structures. Besides, looking at the micro observations, it is correct to underline that there are both well-wetted sections and obvious interface gaps between the fibers and matrix. In addition to these findings obtained from SEM observation, the average layer thickness of the sample was also calculated using a Java-based image processing programme (Figure 5A). During the calculation of the average layer thickness, five measurements were made on

the SEM image and the average layer thickness value was found to be $99.798\ \mu\text{m}$ (SD: $2.710\ \mu\text{m}$). This result shows that the part assigned a layer thickness of $100\ \mu\text{m}$ during slicing was produced with 99.798% accuracy and this finding also supports the dimensional accuracy results presented in detail in Figure 3.

To detect the heating response and determine the melting temperature values of the 3D-printed samples, DSC analyses were conducted. In Figure 6A, DSC curves of the produced samples can be monitored and the results show that PA6, PA6/GF, and PA6/CF samples possess melting temperatures of 186.6°C , 216.1°C , and 220°C respectively. These results are compatible with the literature and close to the previous studies.^{46,47} On the other side, fiber addition leads to a rise in the melting

point for the plain PA6 whereas the difference between the glass and carbon added products is highly small.

The term glass transition temperature can be defined as the temperature at which a rigid polymeric sample converts to a more flexible and viscous material. This critical temperature is usually established by way of DMA practices depending on the gradually escalating test temperatures. In this endeavor, to predict approximate glass transition temperatures, $\tan\delta$ outcomes, which were collected after the DMA, were adopted to find the easily distinguished peak values. Figure 6B depicts the $\tan\delta$ peaks of produced composite samples together with altering analyze temperatures and it reflects that the glass transition temperatures are 68°C , 70°C , and 55°C for PA6, PA6/GF, and PA6/CF samples respectively.

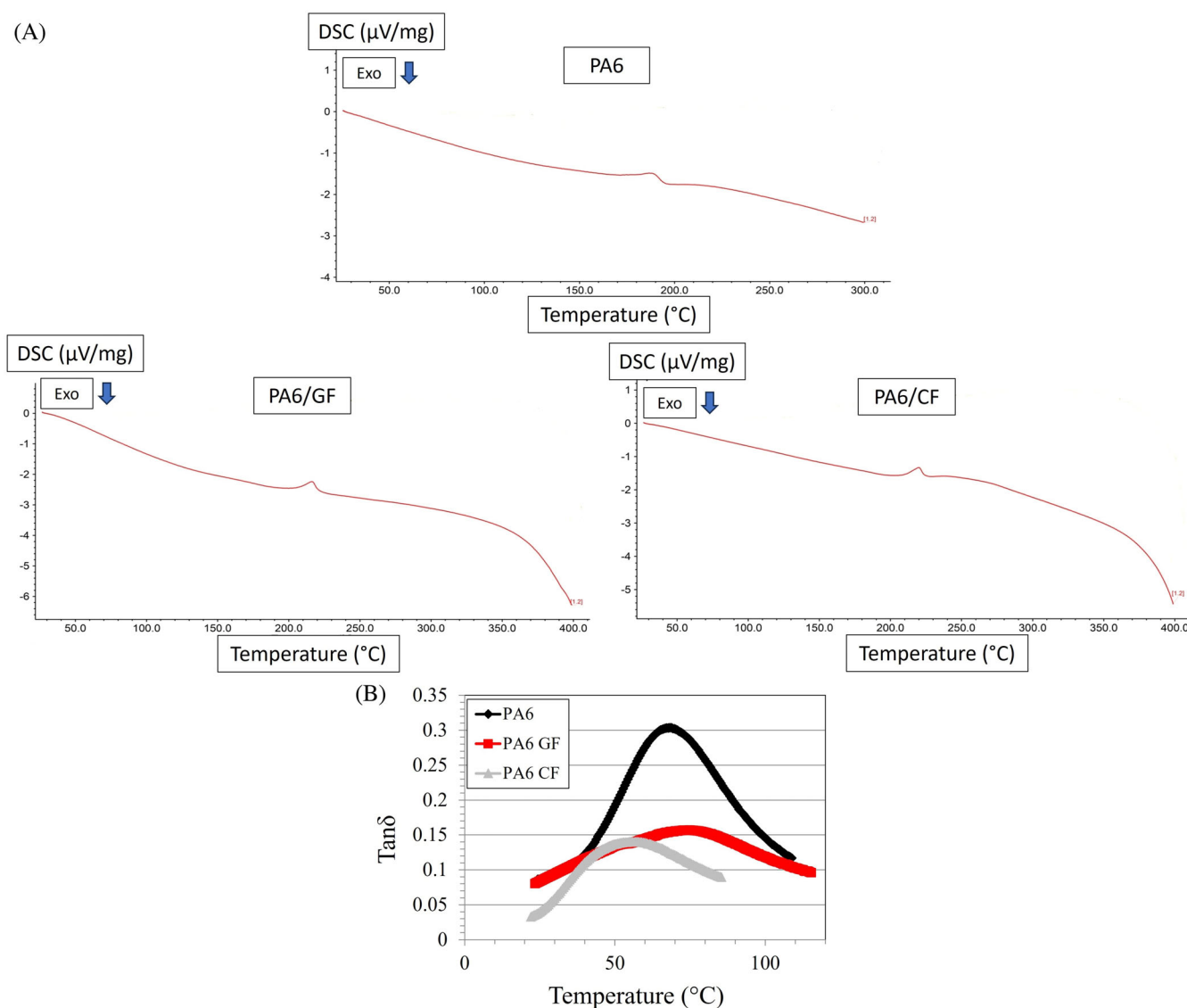


FIGURE 6 Differential scanning calorimeter (DSC) heating curves (A) and dynamic mechanical analyses (DMA)-based $\tan\delta$ graph (B) of the samples.

In the light of the DSC and DMA evaluations, secondary heat-treating temperature levels, or in other saying annealing temperatures, were chosen between the measured glass transition and melting temperatures. Owing to the strong gap-closing and chain-binding potential of the viscous polymer state that is created immediately after the annealing made at higher points than the glass transition, a similar strategy has also been tried in this effort as also happened in certain literature studies.^{39,48} At this point, the temperature difference (70°C was set by the crystallization temperatures of the samples taken into consideration) was kept constant for all types of specimens to provide an identical influence. As a result of this approach, annealing temperatures of 138°C, 140°C, and 125°C were adjusted for PA6, PA6/GF, and PA6/CF samples.

3.3 | Hardness evaluations

Shore D hardness outcomes of the produced polymer composites can be monitored depending upon the altering reinforcement type and heat treatment condition in Figure 7. Before the tests, samples were reconfigured and results were recorded according to the ASTM D2240.

At first glance, the positive influence of the reinforcement material on the hardness behavior is noticed both for normal and heat-treating conditions. This observation can be explained by the relatively higher stiffness of the reinforcement carbon and glass fibers as well as the sufficient bonding between the composite constituents. When the heat-treated and standard samples are compared to each other, heat-treated versions exhibit superior hardness results for all sample types. Thanks to the adhering

stimulative property of the annealing among the neighbors' printing layers, secondary annealing has a promising potential for diminishing fabrication gaps. With the secondary heating and holding time, the average distance between the printing layers that improve the hardness of the samples is reduced. In parallel to this, the highest average Shore D hardness of 80.1 belongs to the heat-treated sample with carbon fibers whereas the lowest result of 77.3 is recorded for the untreated plain PA6 sample. Although the difference between the PA6 and PA6/GF samples is very close, glass fiber-added samples have higher hardness values than the plain versions and their performance can be classified as moderate. After the secondary heat treatment, the average Shore D hardness levels of PA6/GF samples can reach 78.4. If the general results are analyzed by looking at the literature,^{49–51} both changing trends and attained hardness Shore D levels (75 for PA6, 70–80 for PA6/CF, and 80–85 for 25% glass added PA6) are compatible with the previous efforts made via PA6 composites produced with extrusion and 3D printing.

3.4 | Wear and friction coefficient results

Wear tests carried out in this work were conducted by obeying the standard of ASTM G99 and related samples were printed with dimensions of 10 mm diameter and 20 mm height. As the counter-face, SiC papers with 320 grid size were selected and mounted on the dry sliding test section of the wear setup. Following each test, the counter-face was changed to a new one to check the results in the same way. The volume losses of untreated and heat-treated samples at different load values are shown in Figure 8A,B, respectively. At first glance, it can be observed that the volume loss of the samples increases as the load values increase and the heat treatments applied to the samples reduce their volume loss. Plain PA6 samples show higher volume loss at 5 N, 10 N, and 20 N compared to PA6-CF and PA6-GF. However, this situation changes at 40 N, and the PA6-CF and PA6-GF samples start to lose more volume than the PA6 samples. In fact, based on this finding, it can also be put forward that there is a turning point on the wear behaviors of the samples. This may be due to the deformation of the fiber content in the samples and eventually these fracture-based deformations can lead to the separation of the fibers from the interface areas as the load values approach 40 N. If the other literature efforts are evaluated,^{27,30} the self-lubricative effect of the carbon and glass fibers in the polymer matrix can be considered a probable reason for the low wear rates at lower test forces in comparison with the plain PA6. A minimum average

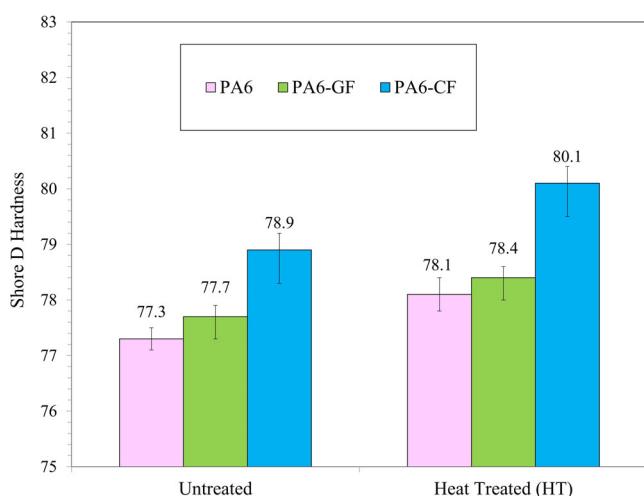


FIGURE 7 Shore D hardness results of the 3D-printed samples.

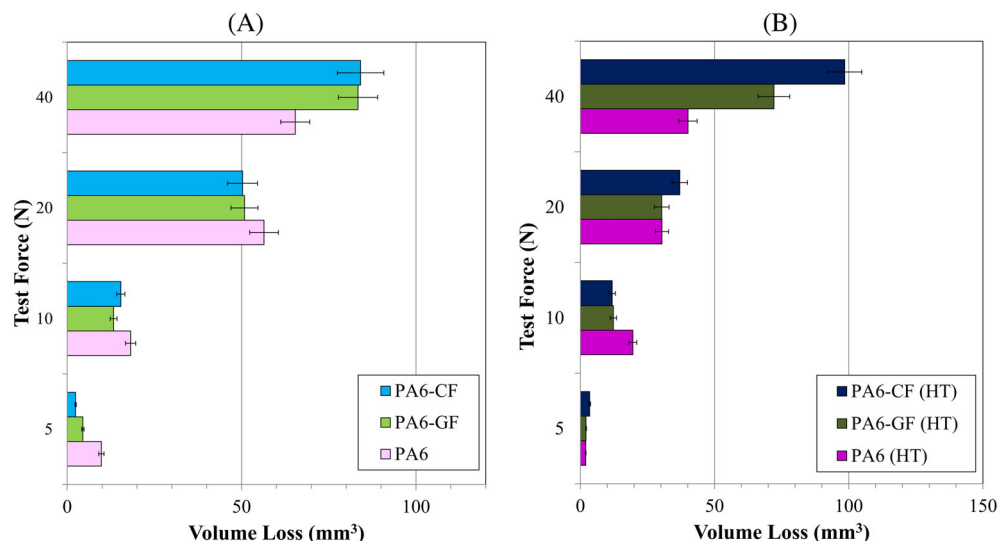


FIGURE 8 Volume loss of the samples under different load conditions, (A) untreated samples, (B) heat treated samples.

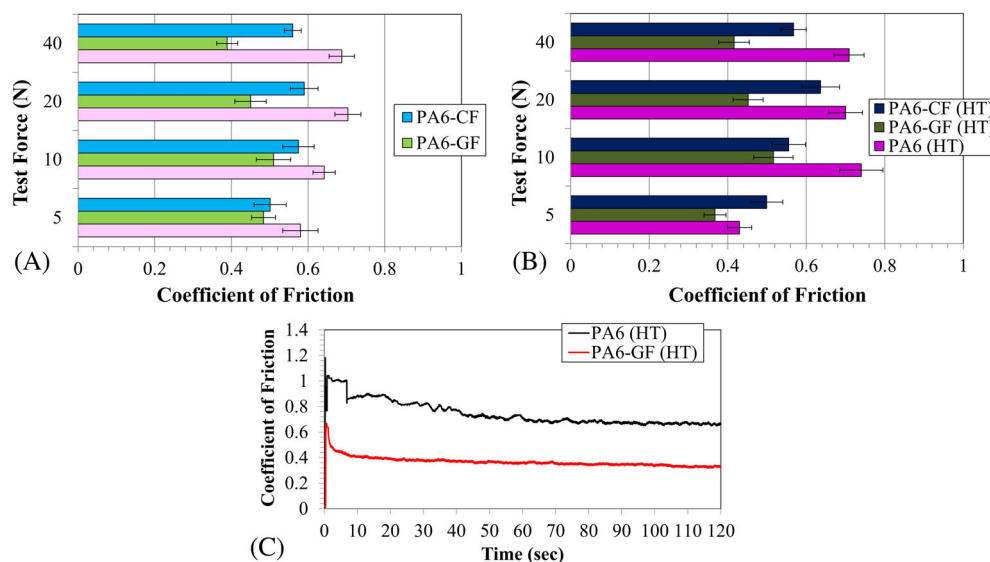
volume loss value of 2.4 mm^3 with a standard deviation (SD) of 0.21 mm^3 was recorded for the PA6-CF sample at 5 N load for unheated conditions. In addition, PA6-GF and PA6 samples showed a volume loss of 4.49 mm^3 (SD: 0.32 mm^3) and 9.8 mm^3 (SD: 0.75 mm^3) respectively at the same load and condition. On the other hand, the effect of fiber resistance to wear was reduced due to the possible high-volume fiber separation of the samples under 40 N load, and a maximum average volume loss of 84.1 mm^3 (SD: 6.65 mm^3) was obtained for the PA6-CF sample. In addition, the average volume loss values of PA-GF and PA6 were in the order of 83.3 mm^3 (SD: 5.62 mm^3) and 65.4 mm^3 (SD: 4.16 mm^3) respectively.

From the experimental results obtained for 5, 10, and 20 N loads, it can be emphasized that the volume loss values of the samples decreased remarkably as the secondary annealing was carried out. For example, at 20 N load, the volume loss of the PA6-CF sample decreased from 50.2 mm^3 (SD: 4.26 mm^3) to 37 mm^3 (SD: 2.89 mm^3). At the same load level, the volume loss of the PA6-CF and PA6 samples decreased from 50.8 mm^3 (SD: 3.88 mm^3) to 30.2 mm^3 (SD: 2.77 mm^3) and from 56.4 mm^3 (SD: 4.12 mm^3) to 30.3 mm^3 (SD: 2.45 mm^3), respectively. Especially for the lower test forces, this kind of material behavior can also be supported by certain literature works^{32,39} delving into secondary annealing. By way of the annealing treatment, the crystallinity degree of the produced polymer goes up owing to the reconfiguration of the primary chains and this circumstance also stimulates the hardness increase in the polymer phase, as well as better mechanical responses. However, the same decreasing trend did not stand out in the 40 N load condition for the PA6-CF sample while PA6 and PA6/GF samples still show the same diminishing inclination. Moreover, fiber-reinforced samples worn more than the plain polymer

samples at 40 N. Obtained consequences are attributed to the combined impact of climbing brittleness of the annealed polymer matrix and partially insufficient matrix/fiber interface bonding for this level of force. On the other side, it is known that the secondary heat treatment processes usually reduce the gaps that occur as a result of the nature of the layer-based additive manufacturing in the sample. Therefore, as seen in Figure 6, minimizing the gaps results in higher hardness values for the heat-treated samples than for the untreated samples. Although the hardness of the material is not the sole factor influencing wear characteristics, it should be taken into account in the interpretations, particularly for the 5, 10, and 20 N loads. In addition, sudden fiber fractures, heterogeneous fiber distribution in the matrix, the inner void gradient in the solid sample bodies, and separations of the fibers at the production stage play a crucial role in the wear mechanisms of the composite specimens.

As can be seen from the comprehensive review of the literature,⁵² coefficient of friction values can vary according to the choice of counter material, normal load, sliding speed and amount of filler. In addition, the wear properties of polymers against polymer or ceramic-based materials, especially when compared to their metal counterparts, have not yet been investigated in depth. Nevertheless, since the industrial usage of the PA6 and PA6/CF/GF composites covers a wide range of areas like sheet manufacturing, coating material options, flame-resistant components for aerospace, and bullet-proof armor layers, the detailed analysis of the friction coefficient values is pretty crucial for the real service performance.⁵³ The coefficient of friction values of the untreated and heat-treated samples subjected to wear tests at different loads are shown in Figure 9A,B, respectively. Herein, it can be asserted that there is no direct relationship between test

FIGURE 9 Variation of coefficient of friction depending on load, (A) untreated samples, (B) heat treated samples, and (C) instant change of the friction coefficient for the highest/lowest results.



load and coefficient of friction. The measured coefficient values go up sometimes together with the rising test load and sometimes drop based on the contact surface condition. If the sample types are checked up with each other, it is realized that plain PA6 samples have the highest coefficient of friction value for each load level and heat treatment condition, followed by PA6-CF and PA6-GF in a row. Among the untreated test samples, the lowest average friction coefficient occurred in the PA6-GF samples with a value of 0.389 (SD: 0.027) under a load of 40 N. Nevertheless, the highest average friction coefficient was measured as 0.704 (SD:0.034) in the PA6 sample at 20 N. In view of heat-treated test samples, the lowest average friction coefficient of 0.367 (SD:0.028) was observed in the PA6-GF sample at 5 N, even though the highest average friction coefficient of 0.74 (SD:0.055) was observed in the PA6 sample under 10 N load (Figure 9C). If these findings are analyzed with other previous efforts, a study by Man et al.⁵⁴ focused on the friction and wear behavior of additively manufactured continuous carbon fiber reinforced PA6 composites and the coefficient of friction value of carbon reinforced PA6 composites was found to be between approximately 0.4 and 0.6 depending on the orientation of the fibers. On the other side, in another study⁵¹ using a high volume of fibers (almost 25%), friction coefficient values dropped to 0.15 levels and this case can be attributed to the positive adhesive effect of the fibers as a result of their perfect lubricative effects. El-Tayeb⁵⁵ reported that the number of fibers in the composite significantly affects the wear behavior of the composite and the friction coefficient value of the composite material can be higher or lower than the matrix material. This finding was also explained in another performance⁵⁶ and was associated with the failure to realize strong bonding.

3.5 | Deformation analyses and wear mechanism

In Figure 10 below, optical microscopic images of unworn and worn PA6 specimens under different loads are presented. According to the surface images, the production layers are more visible in the unworn PA6 specimen. Depending on the level of the plastic deformation, the visibility of the layers decreases from Figure 10A–E (from 5 N load to 40 N load). Along with the deformation-oriented extensions in the layers and adhering effects, jagged surface structures transform into relatively flat surfaces. Due to the high plastic deformation and dominant adhesive wear, all layers disappear when the 40 N load is reached. For altering test forces, similar surface observations were also made for carbon and glass fiber added samples, and the general surface deterioration trend stayed same.

Usually, heat-treated samples exhibited superior wear endurance in comparison with the non-treated samples for all types of samples up to the test force of 40 N. Especially for lower test forces, the positive effect of the secondary annealing was noticed for the volume loss results. To analyze the deformation mechanisms at the lower forces, additional FESEM inspections were carried out. In Figure 11, FESEM outcomes of the heat treated, and non-treated samples can be followed for a low-test force of 10 N. It is seen that secondary debris particles and apparent slot-shaped intrusions are present on the surface of the PA6 and PA6/CF samples. Similar observations including groove-like deformed zones with delamination sections and small debris particles were conducted by other researchers^{25,27} focusing on the wear properties of 3D-printed polymer composites and the decisive influence of the carbon/graphite fiber can be seen on the worn surfaces from these valuable efforts too. However, adhesive

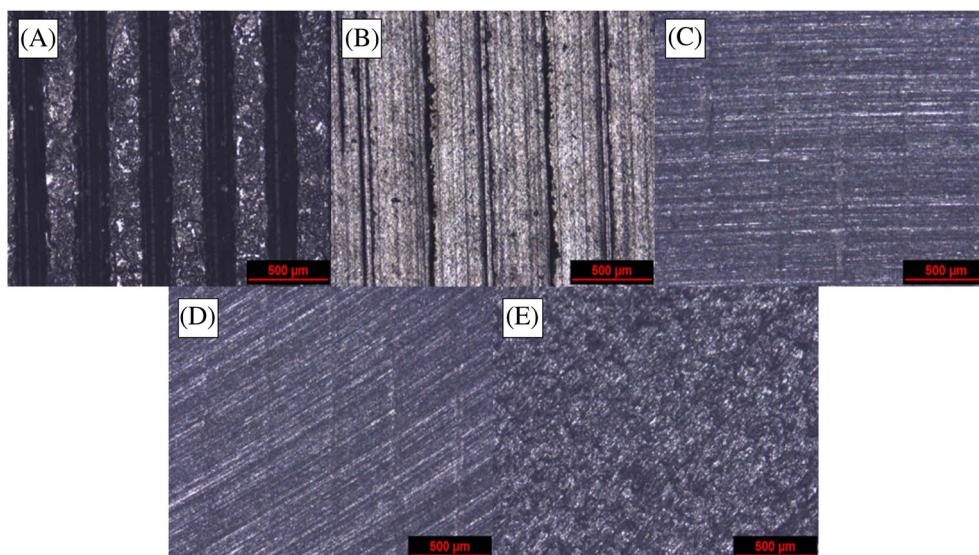


FIGURE 10 Microscopic images of the unworn and worn surface of the polyamide 6 (PA6) samples under different loads; (A) unworn, (B) 5 N, (C) 10 N, (D) 20 N, (E) 40 N.

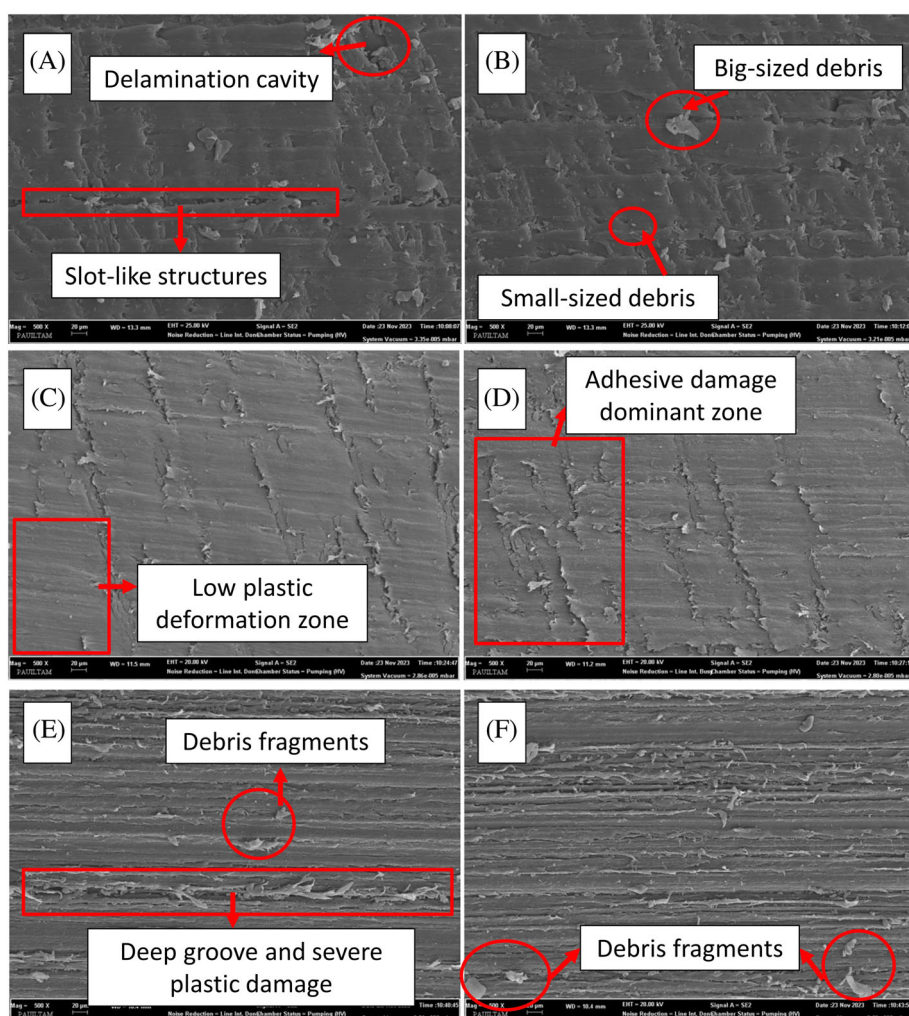


FIGURE 11 Fused emission scanning electron microscope (FESEM) images of the worn surfaces of untreated polyamide 6 (PA6)/carbon fiber (CF) (A), heat-treated PA6/CF (B), untreated PA6/glass fiber (GF) (C), heat-treated PA6/GF (D), untreated PA6 (E), and heat-treated PA6 (F) samples for 10 N.

effects are more widespread on the worn surface of PA6/GF samples and there are no major slot-like structures in the wear zone that trigger an increment in the average surface roughness values of the tested sample.

This case can be supported by the average friction coefficient results and the lowest levels are recorded for the PA6/GF samples (Figure 9). Further, compared to the carbon added versions, debris-like small particles are not

detected for PA6/GF samples and main damage characteristic consists of plastic deformation and adhesion-based wear. When these observations and interpretations are compared to similar scholarly endeavors,⁴⁹ the weak interface binding between the glass fibers and polymer matrix may lead to material transfer to the abrasive counter-face and wave-like adhesion section becoming evident on the PA6/GF part surfaces. The finding is more apparent in this article due to the relatively lower volume ratio of the fibers compared to the other efforts.⁴⁹ Besides, since the deep slots and jagged surfaces with different-sized debris particles exist on the worn surfaces, PA6 samples have higher friction coefficient values both for heat-treated and untreated samples. Similar to this, PA6/CF samples display adhesion-based deformation with a small number of debris fragments. This situation may stem from the combined impact of the high initial surface roughness (Figure 3B) and relatively high average friction coefficient values of PA6/CF samples.

When the volume loss results are examined in detail, it is true to express that there is a changing trend in the wear behaviors of the 3D-printed samples at 40 N. This circumstance was scrutinized in terms of micro-damage analyses via FESEM observations depicted in Figure 12. Checked up with the others, PA6/CF samples were more remarkable in terms of volume loss alteration. In that sense, the wear-resistant characteristic of PA6/CF samples began to drop from the point of the 20 N test force. On the other side, a similar diminishment inclination was also observed for PA6/GF samples at 40 N. From Figure 12A,B, some deep cavities and debris-like small fragments can be monitored for PA6/CF samples. For PA6/CF samples, the number of bigger debris rich sections are more than the as-printed versions and the cavity dimensions were bigger for the heat-treated samples. This finding can be explained by the low fracture toughness of the heat-treated samples and insufficient ductility of the reinforcement fibers that become more decisive at higher

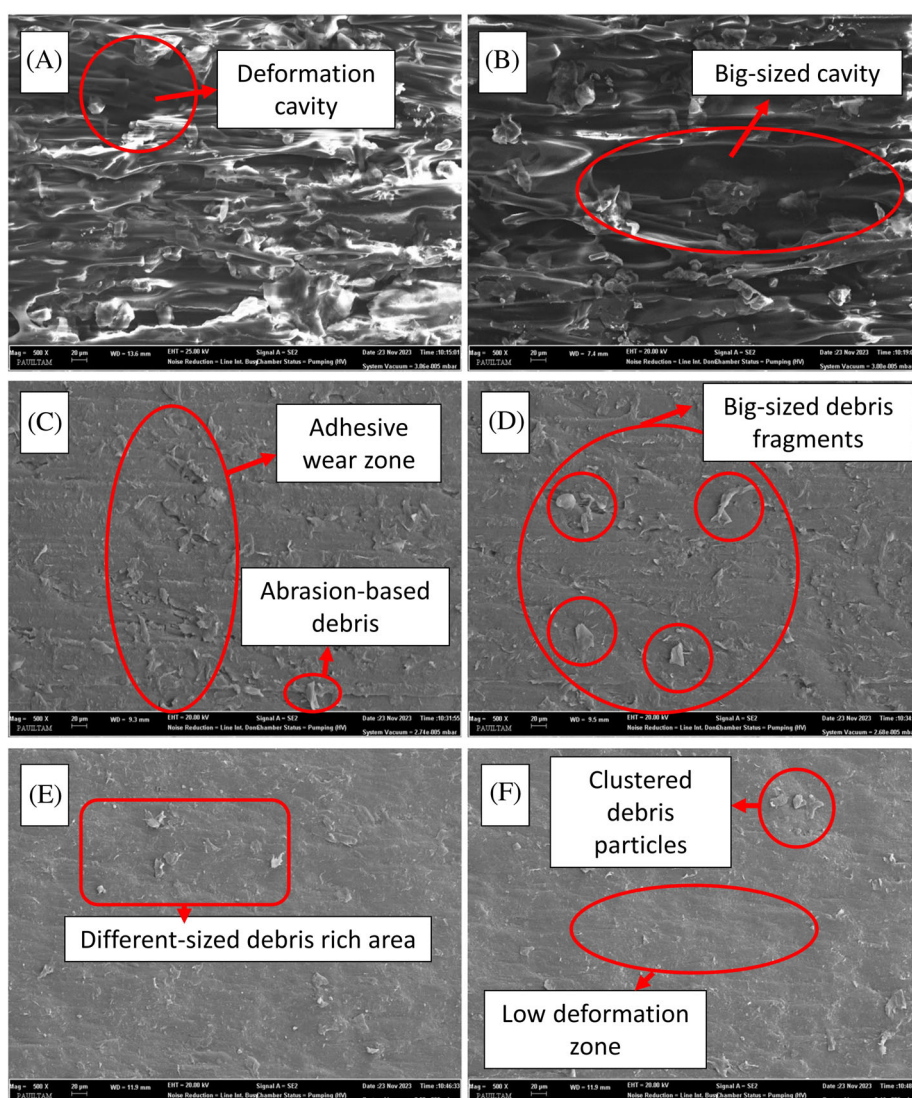


FIGURE 12 Fused emission scanning electron microscope (FESEM) images of the worn surfaces of untreated polyamide 6 (PA6)/carbon fiber (CF) (A), heat-treated PA6/CF (B), untreated PA6/glass fiber (GF) (C), heat-treated PA6/GF (D), untreated PA6 (E), and heat-treated PA6 (F) samples for 40 N.

load levels. Additionally, separated sections of the samples may include the load-bearing and wear-resisted carbon fiber particles. For heat-treated samples, this phenomenon is more effective because of the strengthening effect of the secondary annealing between the printed layers that causes bad elongation ability in the composite body. It is noted in the literature¹⁰ that after the thermal treatment, the distance between the printing layers diminishes as a result of molecular activity controlled by the temperature and time, so present or forming cracks can easily propagate in the polymeric matrix and result in unwanted rapid breakage together with deformation cavities. As for the PA6/GF samples, different from the case at lower forces, small, separated particles are distinguished on the contact surfaces. This observation points out the mixed-type deformation mechanism consisting of adhesive and abrasive style wear. From the point of tribology, Parallel observations were also made by Zaghloul et al.⁴⁹ for PA6/GF samples. It was seen that the combined effect of the abrasive wear, pit forming, and debris-based plastic deformation was decisive in the wear mechanism of the samples. Also, if the initial surface conditions of the fabricated test samples are taken into account, particle breakage tendency of glass and carbon-added versions can be seen as riskier at the higher test levels. Different from the outcome that occurred at lower test forces, PA6 samples exhibit almost no slot-like surface morphology and adhesive effects are more dominant on the contact surfaces. Owing to the contact surface adhesion, smooth surfaces form, and the contact condition converts to polymer/polymer with the rising test force for PA6 samples.

4 | CONCLUSION

As a result of this article that focuses on the effect of the secondary annealing, reinforcement type, and test force on the dry sliding wear performance of additively manufactured PA6 samples, it is correct to assert that secondary heat treatment can be applied to the 3D-printed PA6 composites to obtain superior wear endurance, in particular for the lower test forces. Additionally, compared to the PA6 and PA6/CF, glass fibers exhibit better wear performance up to 40 N and lead to lower friction coefficient levels for all test force levels. At this point, if friction-based problems like thermal softening, partial chemical bonding between the contact surfaces, and unwanted chemical reaction are critical for the specific application, PA6/GF samples are recommended. On the other hand, due to the high initial roughness and low fracture toughness of the fibers, PA6/CF samples become more brittle and sensitive at higher wear test forces. At lower test forces, composite structures come into the forefront and

fiber-reinforced samples usually show better wear resistance in comparison with plain PA6 samples. As for the deformation style, especially at the lower test forces, there are debris fragments on the contact surfaces of PA6 and PA6/CF samples although the situation is quite different for PA6/GF samples having adhesion zones. As the test force escalates and reaches up to 40 N, brittle nature of the carbon and glass fibers becomes more apparent. Furthermore, there is no direct relationship between the friction coefficient and the secondary heat treatment. This case is majorly affected by the dynamic contact surface condition during the sliding tests.

In light of this work, some probable future projects can be considered to understand in depth the wear properties of the short fiber-added polymers made through additive manufacturing. Herein, the combined effect of the fiber volume ratio and water absorption on the tribological responses of the fabricated components can be investigated. In addition, the influence of the counterface material used in the sliding test can be examined experimentally for the determination of the changing deformation mechanisms and abrasion style. On the other side, machine learning and artificial neural network methods can be used to optimize wear rate and friction coefficient values, which makes further work between different disciplines like software and mechanical engineering sciences.

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DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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